IEI-TIA: Industrial Embodied Intelligence Trustworthy Interpretable Agent for Robotic Long-Horizon and Repetitive Tasks

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***Abstract*—With the rapid advancement of generative artificial intelligence, embodied intelligence is increasingly being integrated into robotic tasks within smart manufacturing. Industrial operations typically feature long-horizon sequences and demand high precision. However, although embodied models, particularly Vision-Language-Action (VLA) models, have shown significant promise in autonomous control with superior generalization capabilities, their inherent closed-box nature lacks the transparency and trustworthiness essential for manufacturing tasks. To address this challenge, we propose the Industrial Embodied Intelligence Trustworthy Interpretable Agent (IEI-TIA), a high-level trustworthy supervisory agent designed for industrial sorting and palletizing tasks. Rather than directly generating low-level control commands, IEI-TIA monitors robotic execution, diagnoses possible failure types at key operational stages, and provides corrective next-step instruction via invoked skills, thereby improving the reliability and interpretability of long-horizon and repetitive robotic tasks. An alignment-enhanced parameter-efficient finetuning method is proposed utilizing paired vision-language data to enable robust trustworthiness identification across various possible robotic failure types in different subtasks in sorting and palletizing. Additionally, we construct a specialized dataset and establish benchmarks for industrial embodied intelligence behavior diagnosis and instructions. Experimental results demonstrate that the fine-tuned trustworthy agent achieves 93.7% diagnosis accuracy for various task failures, representing a substantial improvement of approximately 38% over baseline general-purpose large models. By integrating the agent's step-by-step operational instructions, the success rate of robotic long-horizon and repetitive tasks is improved by 14.0%.**

***Note to Practitioners*—This paper addresses a critical problem in industrial automation, the lack of interpretability and transparency of AI-driven robotic control in multi-stage and long-horizon tasks. In high-precision environments such as sorting or palletizing, AI closed-box models, including Vision-Language-Action (VLA) models, often fail to provide safety guarantees required for manufacturing tasks, as minor execution errors can rapidly propagate and affect the entire sequence negatively. Therefore, we propose IEI-TIA, an autonomous supervisory agent that contributes to the shift from open-loop task execution to closed-loop trustworthy control with capability of providing diagnosis and instructions. Beyond identifying potential robotic failures like off-center grasps or misplaced parts, the agent also provides actionable next-step instructions that allow the embodied intelligence system to self-correct before errors escalate. A key practical advantage is that our industrial alignment-enhanced parameter-efficient finetuning (fine-tuning) method empowers the agent utilizing vision-language models with high failure-identification accuracy 93.7%, and corrective closed-loop logic without the need for retraining. The corresponding dataset and benchmarks offer a standardized framework for practitioners to evaluate and deploy interpretable embodied intelligence within existing automation pipelines. By implementing this agent, when using embodied intelligence models in manufacturing, factories can significantly reduce downtime caused by errors or mistakes made by untrustworthy AI robots, to achieve more robust and autonomous control in long-horizon robotic tasks.**

***Index Terms*—Embodied intelligence, industrial agent, trustworthy AI, vision language model, robotic sorting, robotic palletizing, smart manufacturing.**

I. Introduction

IN THE era of Industry 4.0 [1], [2], [3], industrial smart manufacturing has become a cornerstone of global production efficiency. As automation continues to advance, robotic systems are increasingly expected to move beyond simple repetitive motions and execute complex industrial long-horizon tasks. Unlike general spatial applications, such tasks, including industrial sequential sorting [4], [5] and palletizing, involve stronger sequential dependency and higher execution complexity, and therefore pose more rigorous challenges.

In robotic long-horizon and repetitive tasks [6], the success of each stage fundamentally depends on the correct and precise execution of the previous one. Any early-stage error, such as a part being placed outside the bin or an off-center grasp [7], [8], may propagate through subsequent stages, gradually compromising the integrity, stability, and reliability of the entire operation.

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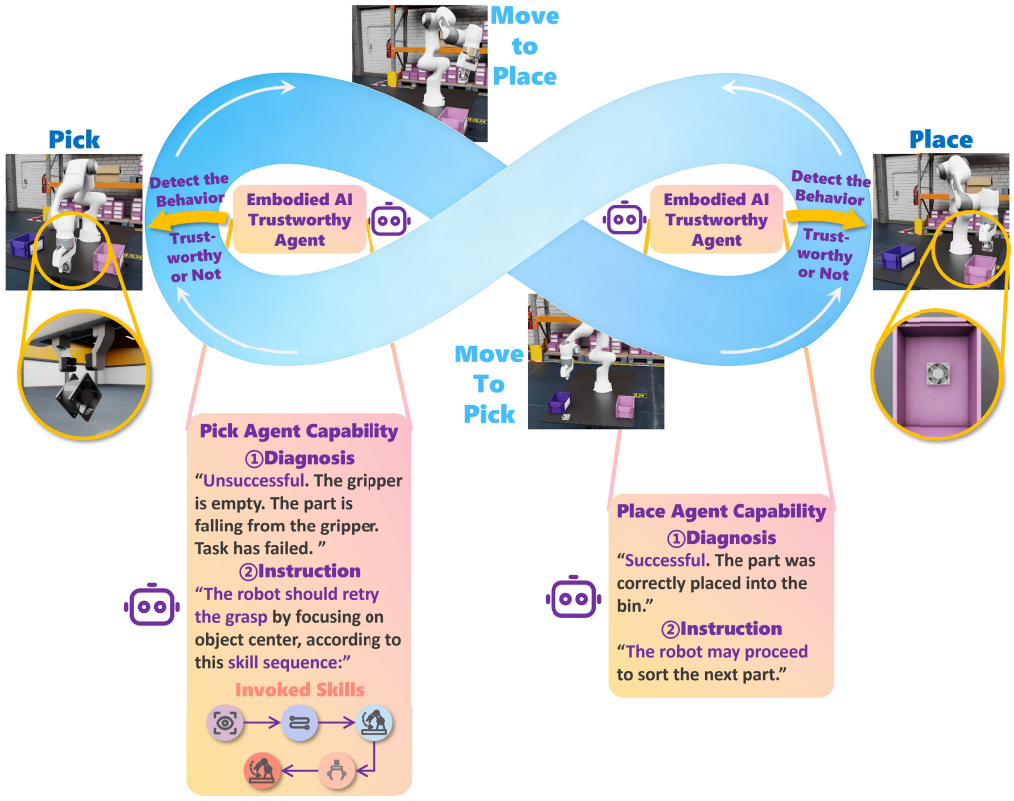


Fig. 1. IEI-TIA in long-horizon and repetitive robotic manipulation tasks. The skills invoked by IEI-TIA in this figure refer to detection, plan, move, grasp and move in sequence. Invoked skills are selected in an industrial embodied tool box and vary according to difference scenarios.

Recent advances in foundation models [9] and embodied intelligence [10], particularly Vision-Language-Action (VLA) and world action models (WAM) [11], [12], have shown strong potential for autonomous robotic control due to their generalization capability [13], [14]. However, existing VLAs and WAMs often operate as closed boxes, lacking the transparency, trustworthiness, and safety required for industrial deployment [15], [16]. For an automated robotic control system to be practically deployed, its decision-making process [17] must be interpretable and its actions must be verifiably reliable [18]. Therefore, to leverage the remarkable generalization and interaction capabilities of multimodal foundation models [19], it is promising to fine-tune [20], [21] foundation models as trustworthy autonomous supervisors for industrial embodied systems.

To address the trustworthiness limitations of AI foundation models in industrial robotic execution, we propose the Industrial Embodied Intelligence Trustworthy Interpretable Agent (IEI-TIA), a high-level supervisory agent for safe automation in robotic sorting and palletizing tasks. Rather than directly replacing the low-level controller, IEI-TIA is designed to evaluate the trustworthiness of robot execution at critical stages, thereby preventing early-stage errors from propagating into subsequent operations. By acting as an autonomous supervisor, IEI-TIA monitors execution outcomes, identifies possible failure types, and further provides actionable and task-specific corrective instruction for the next step based on its diagnostic results. Through this closed-loop mechanism of trustworthiness evaluation, diagnosis, and corrective instruction, the proposed framework transforms conventional open-loop robotic execution into a more robust and trustworthy embodied intelligence system. Experimental results in simulation show that IEI-TIA achieves over 90% robotic failure-type identification accuracy and provides effective stepwise instruction for subsequent behaviors, thereby enhancing robotic trustworthiness, precision, and reliability in long-horizon and repetitive industrial tasks. The main contributions are summarized as follows.

(1) We propose IEI-TIA, an Industrial Embodied Intelligence Trustworthy Interpretable Agent, which realizes closed-loop automation control of embodied agents. By transforming traditional open-loop workflows into a trustworthy closed-loop system, IEI-TIA effectively prevents error propagation in long-horizon and repetitive tasks, such as sorting and palletizing, ensuring continuous trustworthiness and reliability of embodied agents.

(2) To enhance the adaptive diagnosis and corrective instruction capabilities of embodied intelligence systems, we propose an autonomous supervisory mechanism with decoupled failure diagnosis and corrective instruction generation. It enables the agent to evaluate execution outcomes at critical stages, identify potential failure types, and provide task-specific corrective instruction by invoking skills. This mechanism improves closed-loop embodied execution and reduces human intervention.

(3) A novel industrial alignment-enhanced parameter-efficient finetuning (IAE-PEFT) method is proposed for embodied VLMs. To bridge the gap between perception

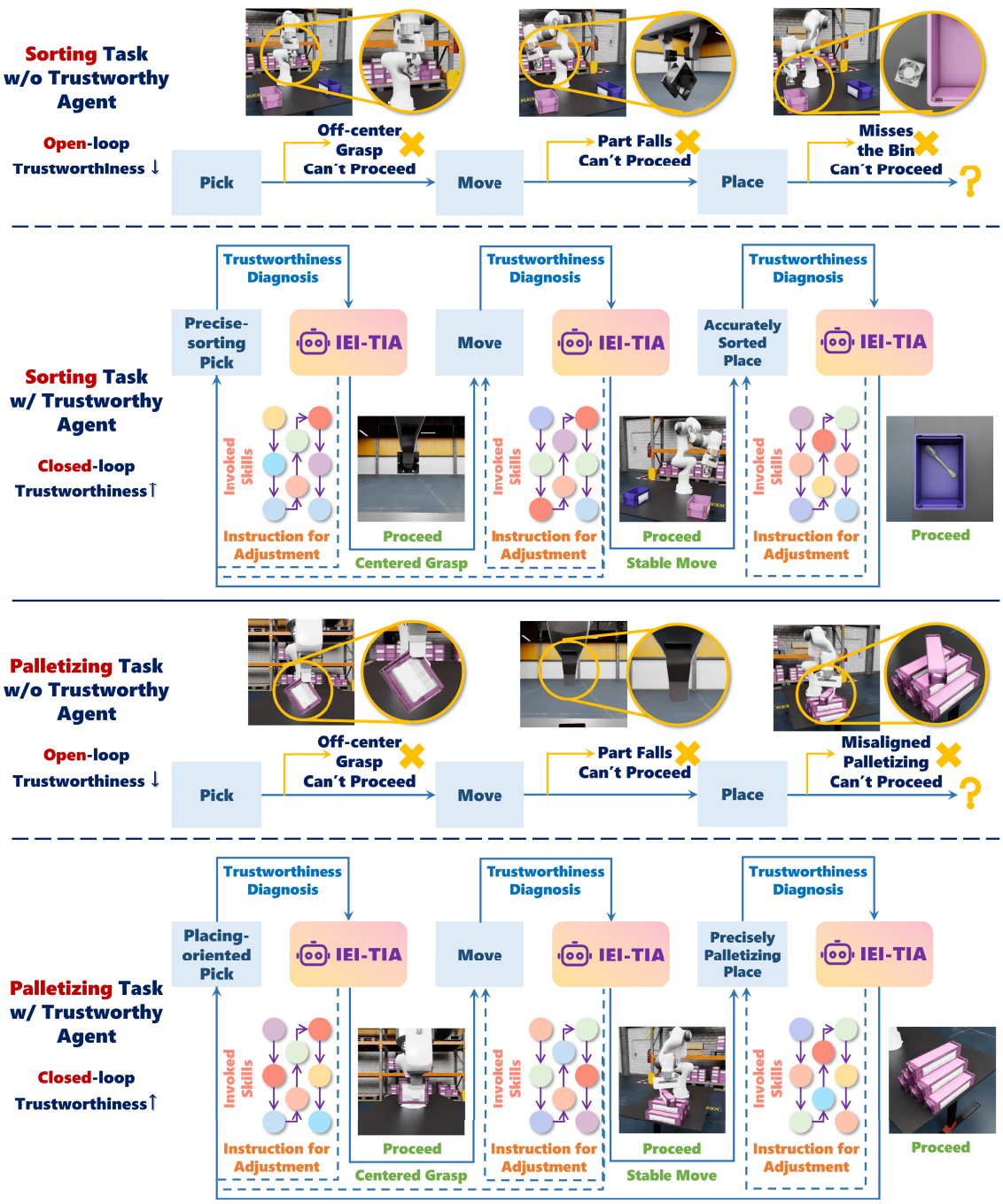


Fig. 2. Open-loop low-trustworthy embodied tasks and closed-loop high-trustworthy embodied tasks with or without IEI-TIA in long-horizon and repetitive robotic manipulation tasks, such as sorting and palletizing.

and control, this method incorporates instruction-aware vision-language alignment, helping mapping visual failure cues directly to corrective robotic next-step instructions. This ensures the agent simultaneously outputs trustworthy diagnostics and high-fidelity next-step instructions with minimal computational overhead.

(4) We construct an Industrial Trustworthy Detection for Failure Cases (ITDFC) dataset and benchmark. The dataset and benchmark are specifically used for agent to realize possible failure type identification of embodied intelligence in industrial long-horizon and repetitive tasks, including sorting and palletizing scenarios.

II. Industrial Embodied Intelligence Trustworthy Interpretable Agent

To overcome the closed-box limitations of embodied intelligence, represented by VLA models, and enhance interpretability of robotic behaviors in long-horizon and repetitive industrial tasks, we propose the IEI-TIA. This agent transforms traditional open-loop robotic control into a robust [22], self-correcting and closed-loop trustworthy system. The IEI-TIA functions as a high-level cognitive layer that monitors and guides the robot's physical execution as a supervisor for several core phases such as pick, move, and place. Unlike standard automation, IEI-TIA inserts a trustworthiness evaluation step

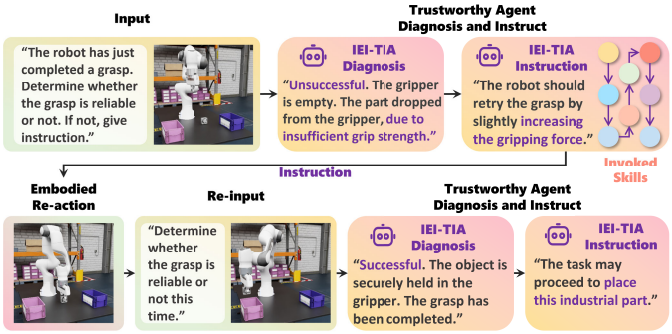


Fig. 3. Input and output of industrial embodied intelligence trustworthy interpretable pick agent. The instruction output by IEI-TIA leads robots to higher execution accuracy of pick sub-task for subsequent part place in long-horizon and repetitive tasks.

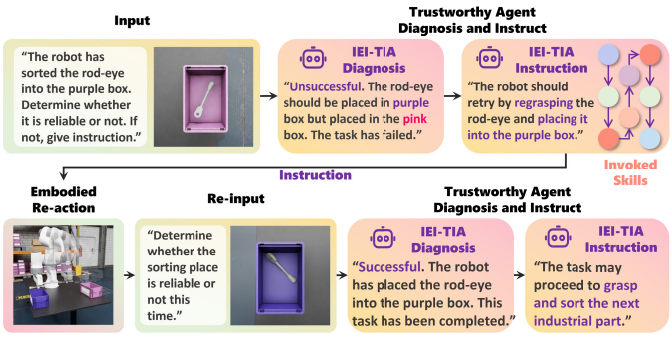


Fig. 4. Input and output of industrial embodied intelligence trustworthy interpretable sorting place agent. The instruction output by IEI-TIA leads robots to higher execution accuracy of sorting place sub-task for next part sorting in long-horizon and repetitive tasks.

after each key primitive action. During this phase, the agent analyzes real-time visual feedback to determine if the behavior is trustworthy or unsuccessful. If a failure is detected, such as an empty gripper or a falling part, the agent generates corrective instruction, such as instructing the robot to retry the grasp by focusing on the center. Execution proceeds to the next stage of the long-horizon and repetitive sequence only after the behavior is verified as successful.

A. Overview of IEI-TIA

IEI-TIA is designed as a two-part trustworthy supervisory architecture, consisting of an *Embodied Diagnostic Agent* and an *Embodied Instruction Agent*, and is intended to work in coordination with a downstream VLA or other embodied execution model. Rather than directly replacing the low-level controller, IEI-TIA operates as a high-level cognitive supervisory layer on top of the embodied policy. In each execution cycle, the underlying VLA or embodied model first performs a primitive action, such as grasping, moving, or placing, according to the current task instruction. The resulting multimodal observation, together with the task context, is then fed into the Embodied Diagnostic Agent, which determines whether the current execution is successful or failed, localizes the failed execution stage when an abnormality occurs, and identifies the corresponding failure type. In this way, the Embodied Diagnostic Agent serves as the first decision layer of the

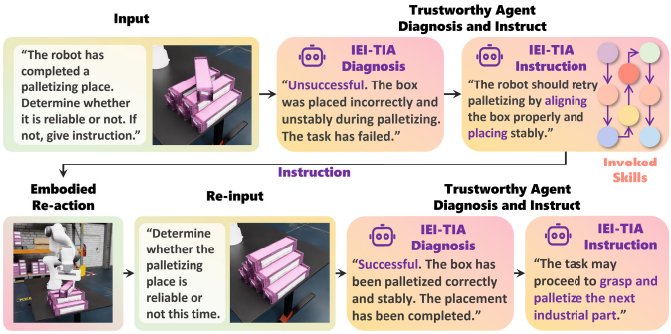


Fig. 5. Input and output of industrial embodied intelligence trustworthy interpretable palletizing place agent. The instruction output by IEI-TIA leads robots to higher execution accuracy of palletizing place sub-task for next part palletizing in long-horizon and repetitive tasks.

framework, transforming multimodal execution feedback into structured diagnostic information for subsequent corrective instruction. Its output includes execution status, failure-stage localization, and failure-type identification.

Based on the diagnostic output and the original task instruction, the Embodied Instruction Agent further generates correction-oriented instruction for subsequent execution. Specifically, it first produces human-interpretable high-level corrective instructions to support operator understanding and human-agent interaction. These high-level instructions are then decomposed into atomic task commands that can guide downstream corrective behaviors, such as re-localization, re-grasping, pose adjustment, force adaptation, or re-place. The resulting corrective commands are subsequently passed to the downstream embodied execution model, skill library, or motion planner, which converts them into executable robot actions. Therefore, the Embodied Instruction Agent does not directly output low-level control signals, but instead acts as a decision-making layer that bridges structured diagnosis and executable corrective instruction.

Through this interaction mechanism, IEI-TIA forms a closed-loop supervisory framework with VLA or other embodied intelligence models. Taking multimodal observations and task instructions as input, it evaluates execution outcomes through the diagnosis agent and, when needed, generates correction-oriented outputs through the instruction agent, including high-level corrective instruction and atomic action instructions for downstream execution. These outputs are then grounded by embodied policies, skill libraries, or motion planners to perform adjusted behaviors. The system proceeds to the next stage only after the current step is verified as successful. In this way, IEI-TIA provides trustworthy monitoring, diagnosis, and correction throughout long-horizon industrial robotic tasks.

B. Decoupled Supervisory Mechanism for Diagnosis–Instruction Consistency

To ensure semantic and logical consistency between diagnostic feedback and subsequent corrective instruction, IEI-TIA adopts a decoupled supervisory mechanism that explicitly separates failure diagnosis from instruction generation. Instead of jointly modeling diagnosis and corrective instruction within a single generation process, the proposed framework

decomposes the closed-loop supervision pipeline into two functionally distinct but tightly coordinated stages. The first stage is handled by the Embodied Diagnostic Agent, which analyzes the current multimodal observation together with the task instruction and produces structured diagnostic results, including execution success/failure assessment, failure-stage localization, and failure-type identification. These outputs provide an explicit and structured description of the current execution state, thereby serving as the semantic basis for subsequent correction-oriented reasoning.

Conditioned on the diagnostic output and the original task instruction, the Embodied Instruction Agent generates corrective instruction for the subsequent execution stage. Specifically, the diagnostic result constrains the instruction generation process along three dimensions: whether correction is necessary, at which execution stage the correction should be applied, and what type of corrective action is appropriate for the identified failure. Accordingly, instruction generation is no longer driven by an unconstrained interpretation of raw observations, but is grounded in an explicit characterization of the current failure state. The Embodied Instruction Agent first produces human-interpretable high-level corrective instruction and may further decompose it into atomic task commands for downstream corrective execution. Through this diagnosis-conditioned generation process, correction is confined to a semantically coherent and logically feasible action space, thereby reducing the risk of inconsistency between failure identification and corrective response.

The proposed decoupling mechanism offers two main advantages. First, it improves diagnostic reliability by allowing the diagnosis module to focus on trustworthiness assessment and failure understanding, which is particularly important for recognizing subtle but safety-critical abnormal states in industrial scenarios. Second, it enables more flexible and targeted corrective instruction. Because the instruction module is conditioned on structured diagnostic feedback, it can generate context-aware corrective instructions while maintaining consistency with the diagnosis result. As a result, the overall framework better balances diagnostic accuracy, instruction flexibility, and execution safety in long-horizon industrial robotic tasks.

III. Industrial Embodied VLM Alignment-Enhanced Parameter-Efficient Finetuning Method for Trustworthy Detection

To enable the proposed agent to effectively identify [23], [24] and rectify failures in complex industrial environments, we develop the Alignment-enhanced Parameter-Efficient Finetuning. As detailed in algorithm, this method bridges the gap between high-level visual hidden states and actionable textual reasoning through a dual-objective optimization strategy that ensures both cross-modal semantic alignment and generative accuracy.

A. Architectural Design and Parameter-Efficient Finetuning

The core architecture of our embodied Vision-Language Model (VLM) integrates a frozen Vision Encoder (e.g., ViT) [25] and a Large Language Model (LLM) backbone. We adopt a Parameter-Efficient Finetuning (PEFT) strategy. Specifically,

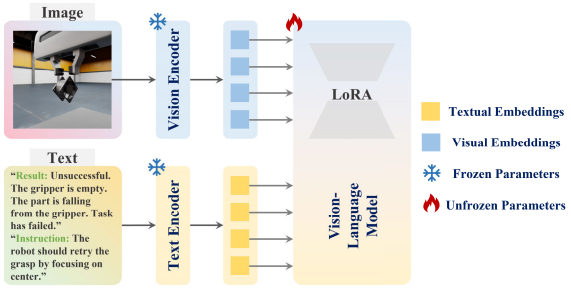


Fig. 6. Illustration of IEI-TIA parameter-efficient finetuning. Architecture of the IAE-PEFT module. The snowflake icon denotes frozen parameters Θ_{fixed}, while the flame icon denotes the trainable parameter updates $\Delta\Theta$, specifically optimized for industrial agent tasks.

the parameters of the Vision Encoder and the Text Encoder are frozen (indicated by the snowflake icon) to preserve robust feature extraction and prevent catastrophic forgetting.

To achieve task-specific adaptation, we integrate Low-Rank Adaptation (LoRA) modules [26] across the entire VLM architecture. Unlike standard LLM finetuning [27], our approach unfreezes only the LoRA parameters $\Delta\Theta$ and a specialized Alignment Head $\mathcal{A}$ (indicated by the flame icon). This configuration allows the model to learn the intricate mapping between visual irregularities, such as a falling workpiece, and industrial semantics without the computational overhead of full-parameter updates.

B. Alignment-Enhanced Training Objective

The training process follows a multi-step optimization logic. The framework simultaneously optimizes for two distinct losses:

- **Latent Alignment Loss** ($\mathcal{L}_{align}$): Using a Unified Cross-Modal Embedding Projection, the model projects visual representations into a shared latent space. The Alignment Head $\mathcal{A}$ is supervised by a Cosine Similarity loss against textual semantic anchors H_{text_anchor}, ensuring that the visual 'hidden states' of a failure are mathematically proximal to the textual description of that failure:

$$\mathcal{L}_{align} = 1 - \text{CosineSim}(\mathcal{A}(H_{vis_rep}), H_{text_anchor}) \quad (1)$$

- **Supervised Finetuning Loss** ($\mathcal{L}_{gen}$): The model performs end-to-end instruction response generation. Given a prompt t_{prompt} and visual input v, the model generates a predicted response Y_{pred}, optimized using Cross-Entropy loss:

$$\mathcal{L}_{gen} = \text{CrossEntropy}(Y_{pred}, t_{target}) \quad (2)$$

C. Trustworthiness-Oriented Output Generation

IEI-TIA supports closed-loop trustworthy execution through iterative diagnosis, corrective instruction, and re-verification. Given the current observation and task instruction, the system produces structured outputs from the Embodied Diagnostic Agent and the Embodied Instruction Agent. The Embodied Diagnostic Agent first determines whether the current execution is successful or failed. If a failure is detected, the Embodied Instruction Agent further generates correction-oriented outputs at two levels: high-level corrective instruction

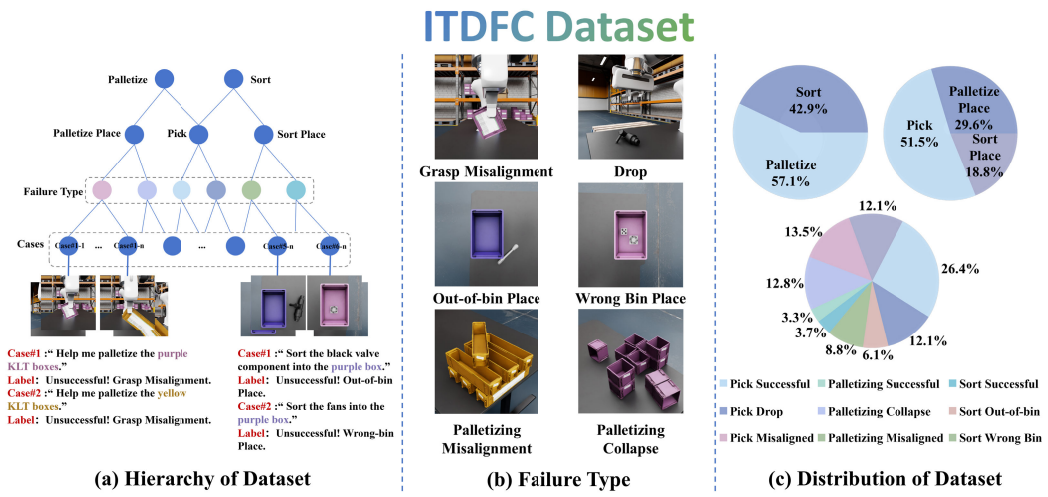


Fig. 7. Overview of industrial trustworthy detection for failure cases (ITDFC) dataset.

and low-level corrective instruction decomposition for downstream execution.

For instance, after a grasp action, the diagnostic output may indicate: *"Unsuccessful. The gripper is empty. The part dropped from the gripper."* Based on this result, the instruction output may provide high-level instruction such as retrying the grasp with improved centering and gripping force, together with low-level instruction decomposition including object detection, grasp planning, trajectory generation, gripper alignment, force adjustment, grasp verification, and stable lifting. After instruction, the updated observation is re-input for a new diagnosis. Execution proceeds to the next stage only when the diagnostic result confirms success, e.g., *"Successful. The object is securely held in the gripper. The task may proceed."*

Importantly, the generated instruction is not a fixed template associated with a single failure label. Instead, it is conditioned on the diagnosed failure state, execution stage, and task context, so that different failures and operational scenarios can trigger different corrective instructions and low-level decompositions. This enables the instruction module to remain both semantically consistent and operationally adaptive. Through this iterative output mechanism, IEI-TIA transforms conventional open-loop execution into a trustworthy closed-loop process.

IV. Industrial Trustworthy Detection for Failure Cases Dataset

To support trustworthy diagnosis and correction-oriented supervision in industrial embodied tasks, we construct the Industrial Trustworthy Detection for Failure Cases (ITDFC) dataset. Unlike conventional manipulation datasets that mainly emphasize successful execution trajectories, ITDFC is specifically designed for failure understanding in long-horizon industrial operations, with a focus on industrial sorting and palletizing. The dataset provides structured multimodal supervision for trustworthiness assessment, failure-stage localization, failure-type identification, and diagnosis-conditioned corrective instruction generation.

Algorithm 1 IEI-TIA Alignment-Enhanced Parameter-Efficient Finetuning

Require: Multimodal dataset $\mathcal{D}$ with (V, T) pairs; Pre-trained VLM model Θ_{vlm} (incorporating vision, connector, and language modules).

Ensure: Optimized VLM parameters $\Delta\Theta$ and Alignment Head $\mathcal{A}$.

Initialize: Load pre-trained VLM weights Θ_{vlm}; Initialize LoRA adapters $\Delta\Theta$ across the entire VLM architecture; Initialize Alignment Head $\mathcal{A}$.
Unfreeze: Enable gradients for LoRA parameters $\Delta\Theta$ and the cross-modal alignment module $\mathcal{A}$.
for each epoch $e = 1$ to E_{max} **do**
  for each batch $(v, t) \in \mathcal{D}$ **do**
  Step 1: Multimodal Latent Representation Extraction
    $H_{vlm} \leftarrow \Theta_{vlm}$
  Step 2: Unified Cross-Modal Embedding Projection
    $H_{vis_rep} \leftarrow \text{Projector}(H_{vlm})$
  Step 3:Textual Semantic Anchor Retrieval
    $H_{text_anchor} \leftarrow \Theta_{vlm}$
  Step 4: Latent Alignment Loss Calculation
    $Z_{align} \leftarrow \mathcal{A}(H_{vis_rep})$
    $\mathcal{L}_{align} \leftarrow 1 - \text{CosineSim}(Z_{align}, H_{text_anchor})$
  Step 5: End-to-End Instruction Response Generation
    $Y_{pred} \leftarrow \Theta_{vlm}.\text{forward}(v, t_{prompt}; \Delta\Theta)$
  Step 6: Supervised Finetuning Loss Calculation
    $\mathcal{L}_{gen} \leftarrow \text{CrossEntropy}(Y_{pred}, t_{target})$
  end for
end for
return Optimized VLM Agent

A. Dataset Construction and Distribution

The ITDFC dataset is constructed through a structured pipeline combining industrial failure analysis, simulation-based scenario reproduction, multimodal data acquisition, and manual annotation. We first investigate representative failure patterns frequently observed in real industrial sorting and

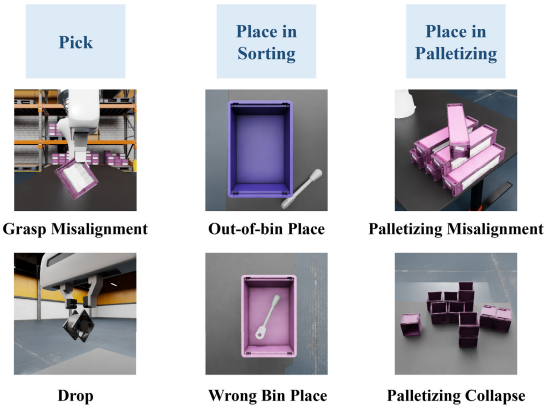


Fig. 8. Typical failure types in pick, place in sorting and place in palletizing subtasks.

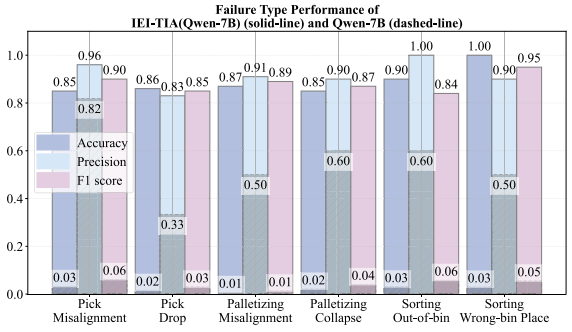


Fig. 9. Failure type identification of Qwen2.5-VL (7B) for sorting and palletizing tasks.

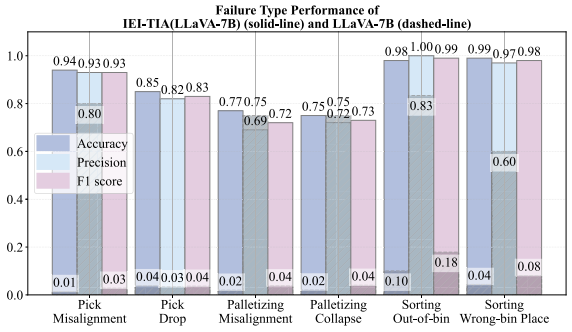


Fig. 10. Failure type identification of LLaVa 1.5 (7B) for sorting and palletizing tasks.

palletizing tasks, as well as common failure types exhibited by current embodied models. These failure cases are then reproduced and diversified in NVIDIA Isaac Sim to ensure controllable, observable, and semantically clear execution outcomes. Based on these simulated industrial scenarios, we collect multimodal samples across different task types, execution stages, and observation viewpoints.

The dataset covers two representative long-horizon industrial tasks, namely sorting and palletizing, and includes both successful and failed executions at key stages such as pick and place. To improve dataset quality, only samples with clear visual evidence, observable abnormal behaviors, and sufficient diagnostic value are retained, while ambiguous or

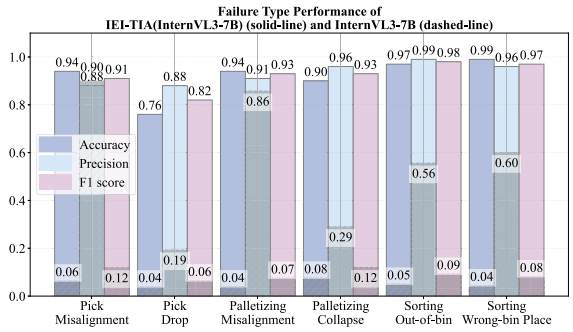


Fig. 11. Failure type identification of InternVL3 (8B) for sorting and palletizing tasks.

weak-evidence cases are filtered or normalized to reduce annotation noise. This strategy ensures that the dataset is not only representative of industrial failure scenarios, but also suitable for reliable diagnosis-oriented learning and evaluation.

B. Dataset Structure and Annotation

To improve transparency and semantic interpretability, ITDFC is organized hierarchically as *scene–task–execution stage–failure type–subtask–execution status*. This hierarchical structure allows each sample to be interpreted not merely as an isolated observation, but as a semantically grounded instance within a long-horizon execution process. Such organization is particularly important for industrial embodied tasks, where failure diagnosis depends not only on visual abnormality itself, but also on its stage-specific context.

All samples are represented in a unified image–text pair format (V, T). Here, V denotes the visual observation of the current execution state, while T provides structured textual annotation describing the associated task semantics. To ensure consistency across tasks and failure categories, we adopt unified annotation criteria at three levels: execution status assessment, subtask-stage localization, and failure-type labeling. Specifically, execution status is determined according to whether the current stage satisfies the physical success condition required for task continuation; subtask localization is assigned according to the stage at which the abnormality first becomes operationally significant; and failure type is labeled based on the primary cause of failure under a predefined taxonomy. Ambiguous or weak-evidence samples are filtered or normalized to reduce annotation noise. This standardized schema explicitly aligns visual evidence with failure semantics, thereby improving both training consistency and evaluation reproducibility.

C. Benchmark

Based on the above hierarchical organization and annotation schema, ITDFC serves as a benchmark for trustworthy supervision in industrial embodied intelligence. Specifically, it supports three core diagnostic tasks: *failure detection*, which determines whether the current execution is successful or failed; *subtask localization*, which identifies the execution stage where the failure occurs; and *failure type identification*, which classifies the corresponding failure category. These benchmark tasks jointly capture the key capabilities required

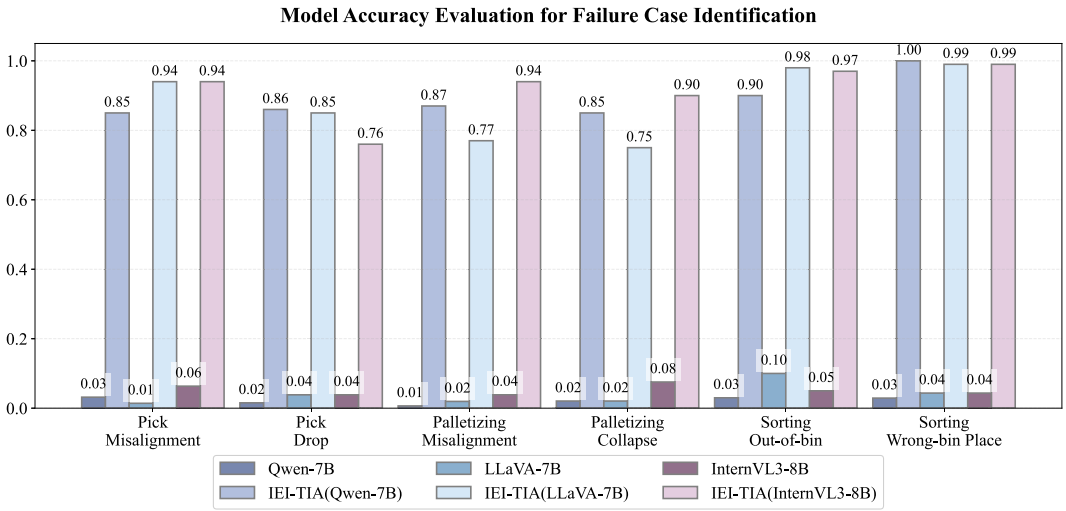


Fig. 12. Comparative experiments of failure type identification for sub-tasks in long-horizon and repetitive and tasks.

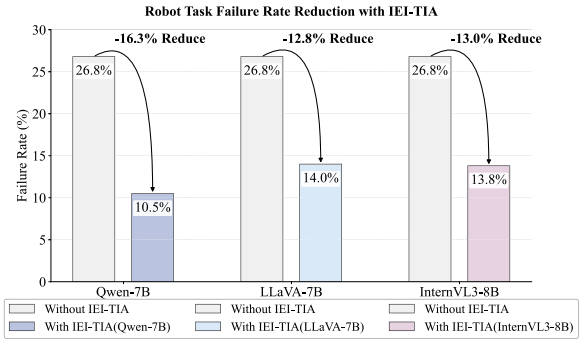


Fig. 13. Robot task failure rate reduction with IEI-TIA.

for closed-loop trustworthy execution, namely state verification, failure understanding, and stage-aware diagnosis.

Beyond diagnosis, the structured annotation of ITDFC also provides supervision for correction-oriented reasoning by linking failure semantics with corresponding corrective feedback. Therefore, the dataset not only supports quantitative evaluation of trustworthy detection performance, but also lays the data foundation for diagnosis-conditioned instruction generation in IEI-TIA. Detailed benchmark settings, evaluation protocols, and experimental results are presented in the subsequent section.

V. Experiments

In this section, we first evaluate five advanced vision-language models, covering both open-source and proprietary systems, on our proposed dataset to establish a rigorous performance baseline. To further examine the effectiveness of the dataset, we fine-tune three representative models, Qwen2.5-VL-7B, LLaVA-1.5-7B, and InternVL-3-8B, yielding the derived variants IEI-TIA(Qwen), IEI-TIA(LLaVA), and IEI-TIA(InternVL), respectively. We then systematically assess these models in industrial scenarios, focusing on their ability to improve failure diagnosis and corrective decision-making. Finally, we investigate the deployment of IEI-TIA in both simulated and real-world robotic environments, demonstrating its capability to support failure correction and ultimately increase task success rates.

A. Experimental Setup

1) Baseline: We evaluate a set of representative vision-language models on our dataset to assess their capability in failure understanding and correction. The benchmark includes both general open-source models (Qwen2.5-VL-7B-Instruct, LLaVA-VL-7B-Instruct, and InternVL-3-8B) and general closed-source models (GPT-4o and Gemini-2.5-Pro). In addition, we report the performance of our fine-tuned variants, denoted as (Ours), which are derived from the corresponding base models. All models are evaluated under the same experimental conditions without additional task-specific prompts or manual tuning.

2) Metrics: We evaluate failure understanding and correction using a set of task-level and diagnostic metrics designed to capture different aspects of model capability.

1) **Failure Detection** measures whether the model can correctly determine if a task execution results in success or failure.
2) **Failure Subtask Localization** evaluates the ability of the model to identify the specific subtask at which a failure occurs within a multi-step process.
3) **Failure Type Identification** assesses whether the model can accurately classify the underlying cause of a detected failure.
4) **Accuracy** denotes the proportion of correctly predicted instances over all evaluated samples for each failure category.
5) **Precision** measures the fraction of predictions assigned to a given failure type that are actually correct.
6) **Recall** quantifies the proportion of true instances of a failure type that are successfully identified by the model.
7) **F1-score** is the harmonic mean of Precision and Recall, providing a balanced view of correctness and completeness in classification.

3) Finetuning With the ITDFC Dataset: We construct a training split from the ITDFC dataset and apply LoRA to

TABLE I

COMPARISON OF MODEL PERFORMANCE ON INDUSTRIAL TRUSTWORTHY DETECTION FOR FAILURE CASES (ITDFC) DATASET. ALL METRICS ARE REPORTED AS ACCURACY

Model	Failure Detection	Failure Subtask Localization	Failure Type Identification	Average
General Open-Source Models				
Qwen2.5-VL-7B-Instruct	23.1%	22.4%	22.2%	22.6%
Llava-VL-7B-Instruct	34.6%	22.9%	22.2%	26.6%
InternVL3-8B	29.0%	27.5%	24.4%	27.0%
General Closed-Source Models				
GPT-4o	57.3%	43.3%	29.0%	43.2%
Gemini-2.5-Pro	52.5%	48.1%	31.3%	44.0%
IEI-TIA with Fine-tuned Models				
IEI-TIA(Qwen2.5-VL-7B-Instruct)	**93.7%**	**93.7%**	**88.7%**	**92.0%**
IEI-TIA(Llava-VL-7B-Instruct)	**93.1%**	**97.3%**	**84.9%**	**91.8%**
IEI-TIA(InternVL3-8B)	**94.4%**	**96.5%**	**90.5%**	**93.8%**

TABLE II

FAILURE TYPE IDENTIFICATION ON INDUSTRIAL TRUSTWORTHY DETECTION FOR FAILURE CASES (ITDFC) DATASET WITH BASE MODELS

Subtask	Type	Qwen-7B			LLaVA-7B			InternVL3-8B		
		Acc.	Prec.	F_1	Acc.	Prec.	F_1	Acc.	Prec.	F_1
Pick	Grasp Misalignment	3.2%	81.8%	6.1%	1.4%	80.0%	2.8%	6.3%	90.0%	11.8%
	Drop	1.5%	33.3%	2.9%	3.8%	3.4%	3.6%	3.8%	19.2%	6.4%
	Average	2.4%	57.6%	4.5%	2.6%	41.7%	3.2%	5.1%	54.6%	9.1%
Place in Palletizing	Palletizing Misalignment	0.6%	50.0%	1.3%	1.9%	75.0%	3.8%	3.9%	85.7%	7.5%
	Palletizing Collapse	2.1%	60.0%	4.0%	2.1%	75.0%	4.0%	7.5%	28.9%	12.0%
	Average	1.4%	55.0%	2.6%	2.0%	75.0%	3.9%	5.7%	57.3%	9.7%
Place in Sorting	Out-of-bin Place	3.0%	60.0%	5.7%	10.0%	83.3%	17.9%	5.0%	55.6%	9.2%
	Wrong-bin Place	2.9%	50.0%	5.5%	4.3%	60.0%	8.1%	4.3%	60.0%	8.1%
	Average	2.9%	55.0%	5.6%	7.2%	71.7%	13.0%	4.7%	57.8%	8.6%
Overall Average		**2.2%**	**55.9%**	**4.2%**	**3.9%**	**62.8%**	**6.7%**	**5.2%**	**56.6%**	**9.2%**

fine-tune Qwen2.5-VL-7B-Instruct, LLaVA-VL-7B-Instruct, and InternVL-3-8B, yielding the adapted variants IEI-TIA(Qwen), IEI-TIA(LLaVA), and IEI-TIA(InternVL), respectively.

B. Main Results

1) Overall Performance of the Models: Overall, the results reveal a pronounced performance gap between general-purpose models and our works. For general open-source models, performance remains modest across the three tasks, with average accuracy ranging from 22.6% to 27.0%, suggesting persistent challenges in reliably detecting failures and identifying their underlying causes. Closed-source general models (GPT-4o and Gemini-2.5-Pro) achieve noticeable gains, reaching average scores around 44%, largely driven by improved failure detection capabilities. In contrast, the IEI-TIA variants consistently attain the highest scores across all metrics, with averages between 78% to 81%. These models demonstrate substantial improvements in both subtask localization and failure type identification, indicating that task-driven adaptation markedly enhances robustness and interpretability. Taken together, these findings underscore that domain-aligned tuning is essential for reliable failure understanding and can outperform even advanced general-purpose LLMs.

2) Overall Model Performance Across Tasks: The experimental evaluation conducted across multiple industrial tasks reveals a substantial disparity between general-purpose vision-language models and the proposed IEI-TIA framework. As detailed in Table I, while general open-source models such as Qwen2.5-VL-7B-Instruct and InternVL3-8B achieve average accuracies of only 22.6% and 27.0% respectively, our fine-tuned IEI-TIA series demonstrates a significant performance leap, reaching an average accuracy range of 91.8% to 93.8%. This performance even exceeds that of prominent closed-source models like GPT-4o (43.2%) and Gemini-1.5-Pro (44.0%), particularly in the 'Failure Detection' metric where our models achieve scores up to 93.8%.

The limitations of base models are further elucidated in Table II, which shows their near-total inability to identify specific industrial failure types. For instance, in the 'Pick' subtask, the LLaVA-7B model records a mere 3% accuracy for 'Empty Grasp' detection. Similarly, in 'Sort Place' operations, base models struggle with place-related errors, with accuracy for 'Place Outside Box' hovering between 1% and 5%. These low scores underscore the necessity of domain-specific fine-tuning for high-stakes industrial applications where error propagation from an initial 'Off-center Grasp' can lead to sequential failures.

The effectiveness of our fine-tuning method is validated by the results in Table III, showing transformative gains across all subtasks. Specifically, for 'Sort Place' tasks, the IEI-TIA(Qwen-7B) model's accuracy for 'Place Outside Box' surged from 3% to 90%. Substantial improvements were also noted in 'Grasp Misalignment' detection, even challenging

TABLE III
FAILURE TYPE IDENTIFICATION ON INDUSTRIAL TRUSTWORTHY DETECTION FOR FAILURE CASES (ITDFC) DATASET WITH IEI-TIA AGENT

Subtask	Type	IEI-TIA(Qwen-7B)			IEI-TIA(LLaVA-7B)			IEI-TIA(InternVL3-8B)		
		Acc.	Prec.	F_1	Acc.	Prec.	F_1	Acc.	Prec.	F_1
Pick	Grasp Misalignment	85.2%	95.7%	90.1%	93.7%	92.7%	93.2%	94.4%	87.6%	90.9%
	Drop	86.2%	83.0%	84.5%	85.4%	81.6%	83.5%	76.2%	88.4%	81.8%
	Average	85.7%	89.3%	87.3%	89.5%	87.1%	88.3%	85.3%	88.0%	86.3%
Place in Palletizing	Palletizing Misalignment	87.0%	90.5%	88.7%	76.6%	68.6%	72.4%	94.2%	91.2%	92.7%
	Palletizing Collapse	84.9%	89.9%	87.3%	74.7%	72.2%	73.4%	89.7%	95.6%	92.6%
	Average	86.0%	90.2%	88.0%	75.6%	70.4%	72.9%	91.9%	93.4%	92.6%
Place in Sorting	Out-of-bin Place	90.0%	100.0%	94.7%	98.0%	100.0%	99.0%	97.0%	99.0%	98.0%
	Wrong-bin Place	100.0%	89.6%	94.5%	98.6%	97.1%	97.8%	98.6%	95.8%	97.1%
	Average	95.0%	94.8%	94.6%	98.3%	98.6%	98.4%	97.8%	97.4%	97.6%
Overall Average		**88.9%**	**91.4%**	**90.0%**	**87.8%**	**85.4%**	**86.5%**	**91.7%**	**92.9%**	**92.2%**

tasks like 'Palletizing Misalignment' in palletizing reached over 95% accuracy with the InternVL-based version. These findings confirm that by utilizing parameter-efficient fine-tuning with paired vision-language data, the IEI-TIA agent successfully transitions from an open-loop system to a trustworthy closed-loop paradigm capable of providing precise corrective instructions.

C. Ablation Study

To verify the contribution of the Embodied Diagnostic Agent in IEI-TIA, we conduct an ablation study based on the Qwen2.5-VL-7B backbone. Specifically, we compare the full framework (*Ours*) with a degraded variant without the Embodied Diagnostic Agent (*No Embodied Diagnostic Agent*). The evaluation is performed from four aspects, including failure detection, failure subtask localization, failure type identification, and robot corrective success rate, so as to examine the role of the diagnosis agent in both failure understanding and downstream correction.

The results show that the full model consistently outperforms the variant without the Embodied Diagnostic Agent across all evaluation metrics. In particular, removing the Embodied Diagnostic Agent leads to clear degradation in error detection, localization, failure type recognition, and robot corrective success rate. These results indicate that the diagnosis agent provides essential structured feedback for subsequent corrective instruction and plays a critical role in the closed-loop reliability of the proposed framework.

D. Discussion

Table I presents a comprehensive comparison of model performance on the ITDFC benchmark across various metrics. The data indicates that general open-source models, such as Qwen2.5-VL-7B-Instruct and InternVL3-8B, exhibit relatively low average accuracy in failure detection and identification, ranging only between 22.6% and 27.0%. While the closed-source model GPT-4o performs slightly better with an average accuracy of 44.0%, it still falls short of industrial requirements. In contrast, the IEI-TIA series models fine-tuned using our proposed method demonstrate significant performance advantages, with failure detection accuracy consistently exceeding 0.90 and overall average accuracy reaching the 0.78 to 0.81 range, substantially outperforming all general-purpose baseline models.

Table II provides a detailed breakdown of the performance of base open-source models across specific industrial subtask failure types, revealing severe limitations in their practical utility. In the Pick subtask, with the exception of InternVL3-8B which shows some capability in identifying an Empty Grasp, models like Qwen-7B and LLaVA-7B show extremely low accuracy in recognizing grasp misalignments and empty grippers, with LLaVA-7B's empty grasp detection falling as low as 3%. In more complex scenarios like Sort Place, all base models perform poorly, with accuracy for identifying failures such as Place Outside Box or Place Wrong Box hovering at a negligible 3% to 5%, rendering them ineffective for real-world long-horizon and repetitive industrial tasks.

Table III illustrates the transformative impact of fine-tuning on the IEI-TIA models, showing a massive leap in their ability to identify various industrial failures. In the Pick subtask, the Qwen-7B version of IEI-TIA improved its grasp misalignment recognition accuracy to 95%.

The results in Table IV suggest a clear correlation between accurate failure type identification and improved execution outcomes. After introducing IEI-TIA-based detection, failure rates decrease consistently across all subtasks and backbone models, indicating that diagnostic feedback effectively supports subsequent correction. This is particularly evident for failure types such as *Drop*, *Grasp Misalignment*, and *Wrong-bin Place*, where correct diagnosis enables more targeted corrective actions, such as grasp adjustment or place replanning, before errors propagate to later stages. More importantly, the observed reduction in failure rates shows that diagnostic feedback does not merely describe abnormal states, but directly facilitates more effective correction. In long-horizon industrial tasks, successful correction depends not only on detecting that a failure has occurred, but also on identifying its cause. Therefore, the stronger reductions achieved by IEI-TIA indicate that more accurate failure type identification provides a better semantic basis for correction and leads to improved overall execution performance.

The ablation results in Table V demonstrate the critical role of the diagnosis agent in IEI-TIA. When the diagnosis agent is removed, performance drops consistently across all evaluation tasks: failure detection decreases from 93.71% to 53.48%, failure subtask localization from 93.66% to 54.36%,

TABLE IV

COMPARISON OF FAILURE RATES BEFORE AND AFTER DETECTION BY THE IEI-TIA AGENT ON SUBTASKS. ALL METRICS ARE REPORTED AS FAILURE RATE. REDUCTION IS CALCULATED AS BEFORE - AFTER (PERCENTAGE POINTS)

Subtask	Type	IEI-TIA(Qwen-7B)			IEI-TIA(LLaVA-7B)			IEI-TIA(InternVL3-8B)		
		Before	After	Reduction	Before	After	Reduction	Before	After	Reduction
Pick	Grasp Misalignment	26%	11%	**15%**	26%	12%	**14%**	26%	11%	**15%**
	Drop	24%	6%	**18%**	24%	10%	**14%**	24%	11%	**13%**
	Average	25%	9%	**16%**	25%	11%	**14%**	25%	11%	**14%**
Place in Palletizing	Palletizing Misalignment	30%	11%	**19%**	30%	17%	**13%**	30%	15%	**15%**
	Palletizing Collapse	18%	9%	**9%**	18%	12%	**6%**	18%	15%	**3%**
	Average	24%	10%	**14%**	24%	15%	**9%**	24%	15%	**9%**
Place in Sorting	Out-of-bin Place	25%	11%	**14%**	25%	15%	**10%**	25%	12%	**13%**
	Wrong-bin Place	38%	15%	**23%**	38%	18%	**20%**	38%	19%	**19%**
	Average	32%	13%	**19%**	32%	17%	**15%**	32%	16%	**16%**
Overall Average		27%	11%	**16%**	27%	14%	**13%**	27%	14%	**13%**

TABLE V

ABLATION STUDY RESULTS FOR INDIVIDUAL COMPONENTS, DIAGNOSIS AGENT IN IEI-TIA. ALL METRICS ARE REPORTED AS ACCURACY (PERCENTAGE POINTS)

Method	**Failure Detection**	**Failure Subtask Localization**	**Failure Type Identification**	**Robot Correction Success Rate**
IEI-TIA	**93.7%**	**93.7%**	**88.7%**	**85.0%**
w/o Diagnosis Agent	53.5%	54.4%	27.7%	40.0%

failure type identification from 88.70% to 27.65%, and robot corrective success rate from 85.00% to 40.00%. These results show that effective correction in industrial long-horizon tasks depends on accurate and structured diagnosis of the current execution state. In particular, the sharp degradation in failure type identification and robot correction success rate indicates that the diagnosis agent provides the essential semantic basis for subsequent corrective instruction and closed-loop trustworthy execution.

VI. CONCLUSION

To address the challenges of error propagation and decision transparency in industrial long-horizon and repetitive tasks, this paper proposes the Industrial Embodied Intelligence Trustworthy Interpretable Agent (IEI-TIA). The proposed agent functions as an autonomous supervisor that realizes robotic possible failure type identification in sequential industrial, such as high-precision sorting and palletizing. The IEI-TIA contributes to the transition of robotic control from open-loop untrustworthy to a closed-loop self-correct system. The Industrial Alignment-Enhanced Parameter-Efficient Finetuning (IAE-PEFT) method enables the agent with capability of real-time failure diagnosis and generation of actionable corrective instructions for subsequent steps. Experimental evaluations demonstrates that IEI-TIA achieves failure recognition accuracy of 93.7%, representing a substantial improvement of approximately 38% over baseline general-purpose large models. Meanwhile, deploying IEI-TIA in sorting and palletizing tasks leads to a 14.0% improvement in robotic task success rates through step-by-step instruction via invoking skills. In future work, it is promising to extend the IEI-TIA to multi-robot collaborative tasks to evaluate its scalability in more complex production lines. In future work, we aim to explore the integration of multi-modal sensor data, such as tactile feedback, to further enhance diagnostic perception ability of agent for more delicate manipulation tasks.

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